

Problem

For a prime number p , define

$$Q(p) = \prod_{k=1}^{p-1} k^{2k-p-1}.$$

Show that $Q(p)$ is an integer.

Source: AMM

§1 Rewriting the Product

We begin by separating positive and negative exponents. Writing out the exponents gives

$$Q(p) = \frac{1^1 2^2 3^3 \cdots (p-1)^{p-1}}{(1 \cdot 2 \cdot 3 \cdots (p-1))^{p+1}} \cdot (1^1 2^2 3^3 \cdots (p-1)^{p-1}).$$

Hence

$$Q(p) = \frac{(1^1 2^2 3^3 \cdots (p-1)^{p-1})^2}{((p-1)!)^{p+1}}.$$

The product

$$1^1 2^2 3^3 \cdots (p-1)^{p-1}$$

can be regrouped as

$$(1 \cdot 2 \cdots (p-1))(2 \cdot 3 \cdots (p-1))(3 \cdot 4 \cdots (p-1)) \cdots (p-1).$$

Therefore

$$1^1 2^2 3^3 \cdots (p-1)^{p-1} = (p-1)! \cdot \frac{(p-1)!}{1!} \cdot \frac{(p-1)!}{2!} \cdots \frac{(p-1)!}{(p-2)!}.$$

Thus

$$1^1 2^2 3^3 \cdots (p-1)^{p-1} = \frac{((p-1)!)^{p-1}}{1! 2! \cdots (p-2)!}.$$

Substituting into the expression for $Q(p)$ yields

$$Q(p) = \frac{((p-1)!)^{p-1}}{(1! 2! \cdots (p-1)!)^2}.$$

§2 Extracting Binomial Coefficients

Since

$$p! = p(p-1)!,$$

we may write

$$Q(p) = \frac{1}{p^{p-1}} \cdot \frac{(p!)^{p-1}}{(1!2! \cdots (p-1)!)^2}.$$

Now distribute the factors of $p!$:

$$\begin{aligned} Q(p) &= \left(\frac{1}{p} \cdot \frac{p!}{1!(p-1)!} \right) \left(\frac{1}{p} \cdot \frac{p!}{2!(p-2)!} \right) \cdots \left(\frac{1}{p} \cdot \frac{p!}{(p-1)!1!} \right) \\ &= \prod_{k=1}^{p-1} \frac{\binom{p}{k}}{p}. \end{aligned}$$

Thus

$$Q(p) = \prod_{k=1}^{p-1} \frac{\binom{p}{k}}{p}.$$

§3 The Crucial Divisibility

Since p is prime and $1 \leq k \leq p-1$, it is well known that

$$p \mid \binom{p}{k}.$$

Indeed,

$$\binom{p}{k} = \frac{p!}{k!(p-k)!},$$

and the factor p appears in the numerator but not in the denominator.

Therefore each quantity

$$\frac{\binom{p}{k}}{p}$$

is an integer.

Since $Q(p)$ is a product of integers, it follows that $Q(p)$ itself is an integer.

§4 The Final Conclusion

We have established the factorization

$$Q(p) = \prod_{k=1}^{p-1} \frac{\binom{p}{k}}{p},$$

and every factor on the right-hand side is an integer because p is prime.

Hence

$$\boxed{Q(p) \in \mathbb{Z}}.$$

This completes the proof.